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Study of Vitamin D Deficiency among the Apparently Healthy Population in Jashore, Bangladesh

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Globally Vitamin D deficiency is a burning issue for the last two decades. Though the very high prevalence of hypovitaminosis D is worldwide even in the South-East Asia but there is very limited data in Bangladesh. This observational study was designed to assess the status of Vitamin D deficiency among the clinically apparent healthy population in the district of Jessore, Bangladesh from January 2017 to June 2017. Both the male and female clinically healthy participants between the age of 10 and 70 years residing more than 3 years in Jashore were included in this study, while those having renal or liver failure or any other chronic diseases and using drugs that could potentially interfere the metabolism of vitamin D were excluded from the study. Here we leveled serum 25(OH)D concentration of ≤20ngm/dl as deficient, >20-30ngm/dl graded as insufficient and >30ngm/dl graded as sufficient. Among 160 participants 43.1% (n=69) were male and 56.9% (n=91) were female. And 102(63.7%) participants had deficient vitamin D, 50(31.3%) had insufficient vitamin D and only 8(5%) had sufficient vitamin D. The mean serum 25(OH)D levels in our study was 18.60±6.59ngm/dl. The mean serum 25(OH)D level in case of male participants was 19.72±7.10ngm/dl whereas in female the corresponding figure was 17.74±6.07gm/dl and no significant difference observed among the man and women (p=0.059). Sun exposure play a vital role in the vitamin D deficiency and we observed only 1-2 hours/day sun exposure was sufficient for normal vitamin D level (p=0.001). In case of women who used veil had significantly lower level of vitamin D (p=<0.001) due to lack of adequate sun exposure. Urban participants who also had significant low level of vitamin D due to the same reason (p=0.009). There was no deficiency observed in farmers and only 5(3.1%) had insufficient vitamin D. No significant difference observed in deficiency of vitamin D among the different age, sex, education level, skin complexion or BMI. But the obese participants had significant (p=0.041) lower vitamin D level. It was concluded with the high rate of vitamin D deficiency in this small scale study conducting in a district of Bangladesh demands a nationwide survey of vitamin D.

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Key words: Vitamin D, Deficiency of vitamin D, 25(OH)D

Introduction

Vitamin D is a prohormone that has a key role in calcium and phosphate balance and bone structure[1]. Vitamin D nutrient functions as a hormone precursor. And had a wide-ranging health effects are supported with the presence of vitamin D receptors in several cell types and tissues of the body (e.g. lymphocytes and monocytes, brain, heart, pancreas, intestine, and placenta)[2]. In addition to its well-established role in the regulation of calcium metabolism, the active form of vitamin D has been shown to have antiproliferative and immunomodulatory effects that are thought to influence the development of several serious conditions, including diabetes, cardiovascular disease, and cancer[3]. Lack of vitamin D manifests as rickets in children and degenerative bone diseases in older people[4].

There is a relationship between vitamin D deficiency and increased risks of developing chronic illnesses, such as type 1 diabetes and cancer as well as cardiovascular diseases in adults[5,6].

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A meta-analysis revealed a potentially deleterious role of low vitamin D in all-cause and cause-specific mortality in primary and secondary prevention cohorts[7]. This finding is significant in public health importance, as the gradual decline in circulating 25-hydroxy vitamin D concentrations reported globally is likely to continue owing to increase in older populations, obesity and lack of adequate sun exposure combined with sunscreen use. Objective of this study is to assess the status of Vitamin D deficiency among the apparently healthy population in the district of Jashore, Bangladesh.

Methods

This observational study was conducted on January to June, 2017 in a teaching hospital named Jashore Medical College & Hospital, Jashore, Bangladesh. The study recruited 160 apparently healthy people who were residing in Jashore for the last 3 years and there was no history of liver, kidney or gastrointestinal disease. Furthermore, pregnant and lactating women, and persons taking vitamin and mineral supplements or drugs (hormone replacement therapy, glucocorticoids, bisphosphonates, teriparatide, antiepileptic drugs) that affect bone metabolism, were excluded from the study. After taking a written informed voluntary consent study subjects were enrolled in this series. Institutional approval was taken from the Superintendent of Jashore Medical College Hospital. After getting the informed written consent, a detailed history and physical examinations were carried out among the index participants. A questionnaire was used to collect information regarding age, sex, religion, residence, educational qualification, occupation, skin complexion, height, weight, dressing habit and daily time spent in the sunny environment. Standing height was measured to the nearest millimeter by using a stadiometer. Participants were weighed in light clothing without footwear. Body weight was measured to the nearest 0.1kg by using an Electronic Weighing Scale (Sinbe Model No, SBS 4414 and made in People's Republic of China). We used the classifications of BMI (weight in kg/height in m^2) recommended by the WHO[8]. The subjects were drawn from five distinct groups: medical professionals, housewife, service holders (government job/private job/business), farmers and students. The usual time per day spent outdoors in daylight hours during the previous month was reported as <30 minutes, 30-60 minutes, 1-2 hours and >2 hours. The participants were classified into four BMI groups as obese (BMI = >30), overweight (BMI = 25-29.9), normal (BMI = 18.5-24.9) and underweight (BMI = <18.5) as established by International Obesity Task Force (IOTF) in the year 2000, based on pooled international data for BMI linked to the adult's cut-off values[8]. In case of female participants, we have categorized their dressing habit into two groups veil (Borkha) and non-veil. Blood samples were collected under basal condition and taking precaution to avoid venostasis. A total of 04ml of venous blood was collected from each participant into Vacutainers (Becton Dickinson Diagnostics). Blood samples were separated by centrifugation at 3000rpm for 10 minutes at room temperature within 2 hours of being drawn and aliquots were stored at 2-4°C until analysis. The estimation of Vitamin D was carried out by Advia Centaur CP/ VitrosEci/Architect-2000Isr/Immulite 2000 Random Access Immunoassay Analyzer/Beckman Coulter Access 2, made in USA. Kit was used for estimation of Vitamin D Beckman Coulter, Lot no.632974, made in USA. The actual cut off level for deficiency of vitamin D is controversial, however, the most experts agree that vitamin D level less than 20ng/ml is considered as deficiency, whereas a level of vitamin D 20-30ng/ml is denoted as insufficient, and a level ≥30ng/ml is the normal value of vitamin D in both children and adults[9]. Preformed structured data collection sheets were used in every selected case. Analysis carried out using SPSS version 23. Categorical data was grouped as % and numbers and mean with standard deviation (SD) measured from continuous data. Independent T-test, one-way analysis of variance (ANOVA) used to extract p value and Games-Howell Post Hoc test used to evaluate deference between different groups.

Results

Of total 160 healthy participants the mean value of vitamin D in this study was 18.60±6.59ngm/dl and 102(63.7%) participants were vitamin D deficient and their mean vitamin D level was 14.81±3.43ngm/dl, 50(31.3%) had insufficient vitamin D level (23.38±2.54ngm/dl) and only 8(05.0%) had sufficient vitamin D level (37.09±2.61ngm/dl) (Table I).

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Table I: Distribution of study participants according to vitamin D level (n=160).

25(OH) D level of study subjects	No. (%)	Mean±SD
Total	160 (100)	18.60±6.59
Deficient (≤20ngm/dl)	102 (63.7)	14.81±3.43
Insufficient (>20ngm/dl-30ng/dl)	050 (31.3)	23.38±2.54
Sufficient (>30ngm/dl)	008 (05.0)	37.09±2.61

Out of total 160 participants 69(43.1%) were man and 91(56.9%) were women with a sex ratio was 0.76:1. Mean age was 44 years with a standard deviation (SD) of 14 years. Highest number of participants was from 3rd to 6th decades. There was no significant difference observed in Vitamin D_3 level among different age groups (p value0.492). It was also observed in deficient and insufficient groups among different age groups. There was significant difference observed in 3rd and 6th decade sufficient groups in vitamin D levels but number of cases in these groups was too low to take these results in consideration. No significant difference observed in vitamin D_3 level among male and female (p value 0.059) (Table II).

Among them 64(40%) subjects were from rural background and rest 96(60%) from urban background. These study participants belonging five occupational groups (physicians, housewife, service and business, farmer and student) were evaluated for vitamin D deficiency. Among them 27(16.9%) were physicians, 61(38.1%) were house wife, 48(30.0%) were of service and business and 13(8.1%) were students (Table III). Study participants had different educational background. Thirty-seven (23.1%) were post graduate and 38(23.8%) completed graduation and rest had primary or secondary educational background. Participants residing in urban area had lower vitamin D_3 level (20.33±7.40ngm/dl) than rural participants (17.44±5.74ngm/dl) significantly (p value 0.009) (Table III).

Table II: Age and sex distribution of study participants according to vitamin D level (n=160).

Characteristics			25(OH) D Level						
				Deficient		Insufficient		Sufficient	
Age in years			Total	≤20ngm/dl		>20ngm/dl-30ng/dl		>30ngm/dl	
Groups	No.(%)	Mean±SD	Mean±SD	No. (%)	Mean±SD	No. (%)	Mean±SD	No. (%)	Mean±SD
11-20	07(04.4)	15±3	18.04±3.83	05(03.1)	16.28±2.90	02(01.3)	22.44±0.14	00(0.0)	-
21-30	23(14.4)	26±3	20.89±8.09	15(09.4)	16.21±1.96	05(03.1)	24.31±3.69	03(1.9)	38.63±0.17
31-40	38(23.8)	36±3	18.36±6.24	23(14.4)	14.10±3.30	14(08.7)	24.40±2.61	01(0.6)	31.90
41-50	37(23.1)	46±3	17.61±5.98	25(15.5)	14.72±3.98	11(06.9)	22.42±2.68	01(0.6)	36.94
51-60	37(23.1)	56±3	18.01±6.52	24(15.0)	14.48±3.94	11(06.9)	22.44±1.24	02(1.3)	35.95±1.34
>60	18(11.2)	67±3	19.60±7.33	10(06.3)	14.59±2.82	07(04.4)	23.83±2.74	01(0.6)	40.10
Total	160(100)	44±14	18.60±6.59	102(63.7)	14.81±3.43	50(31.3)	23.38±2.54	08(05.0)	37.09±2.61
P value			0.492^{ns}		0.466^{ns}		0.263^{ns}		0.01^{s}
Gender	No.(%)								
Male	69(43.1)		19.72±7.10	38(23.8)	14.98±3.64	25(15.6)	22.88±1.77	06(3.8)	36.61±2.90
Female	91(56.9)		17.74±6.07	64(40.0)	14.70±3.33	25(15.6)	23.85±3.10	02(1.3)	38.53±0.03
M: F= 0.76	*P value*		0.059^{ns}		0.700^{ns}		0.181^{ns}		0.409^{ns}

No significant difference observed in Vitamin D_3 Level among different education level (p value 0.589). Farmers had significantly (p value <0.001) higher Vitamin D_3 level (30.49±7.38ngm/dl) than others (Table III).

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Table III: Vitamin D level of study participants according to locality, different educational level and occupation (n=160)

Characteristics		25(OH) D Level						
			Deficient		Insufficient		Sufficient	
Locality		Total	≤20ngm/dl		>20ngm/dl-30ng/dl		>30ngm/dl	
Rural	64(40.0)	20.33±7.40	32(20.0)	14.64±3.22	26(16.3)	23.58±2.68	6(3.8)	36.61±2.90
Urban	96(60.0)	17.44±5.74	70(43.8)	14.88±3.55	24(15.0)	23.14±2.42	2(1.3)	38.53±0.03
P value		0.009^{s}		0.739^{ns}		0.542^{ns}		0.409^{ns}
Education								
Primary	25(15.6)	20.18±5.49	13(08.1)	16.51±3.27	11(06.9)	23.01±2.31	01(0.6)	36.90
Secondary	60(37.5)	18.59±7.10	39(24.4)	14.55±3.57	17(10.6)	23.78±3.03	04(2.5)	35.99±3.44
Graduation	38(23.8)	18.04±5.54	26(16.3)	15.26±3.10	11(06.9)	22.72±1.47	01(0.6)	38.83
Post-graduation	37(23.1)	18.10±7.12	24(15.0)	13.81±3.41	11(06.9)	23.74±2.54	02(1.3)	38.53±0.03
P value		0.589^{ns}		0.114^{ns}		0.665^{ns}		0.729^{ns}
Occupation								
Physician	27(16.9)	17.54±3.57	19(11.9)	15.60±2.09	08(05.0)	22.16±1.20	00(0.0)	-
House wife	61(38.1)	17.17±5.56	43(26.9)	14.41±3.68	18(11.3)	23.77±3.23	00(0.0)	-
Service & Business	48(30.0)	17.40±5.61	33(20.6)	14.41±3.73	15(09.4)	23.96±2.59	00(0.0)	-
Farmer	11(06.9)	30.49±7.38	00(00.0)	-	05(03.1)	23.15±1.46	06(3.8)	36.61±2.90
Student	13(08.1)	21.83±8.01	07(04.4)	16.93±2.68	04(02.5)	22.05±0.47	02(1.3)	38.53±0.03
P value		$<0.001^{s}$		0.196^{ns}		0.393^{ns}		0.409^{ns}

We divided the participants into four groups according to body mass index (BMI). Nine (5.6%) participants had BMI less than 18.5 that is underweight, 57 (35.6%) participants had standard BMI, 72 (45.0%) participants were overweight and rest 22 (13.8%) were obese. Participants had different skin complexion and they were black (35; 21.9%), brown (59; 36.9%) and fair (66; 41.2%). No significant difference observed in Vitamin D_3 level among different BMI groups but obese deficient subjects had significant (p value 0.041) lower vitamin D_3 level. In this study we also found no significant differencein Vitamin D_3 level among subjects with different skin complexion (Table IV).

Table IV: Vitamin D level of study participants according to BMI and skin complexion (n=160)

Characteristics		25(OH) D Level						
			Deficient		Insufficient		Sufficient	
BMI		Total	≤20ngm/dl		>20ngm/dl-30ng/dl		>30ngm/dl	
<18.5	09(05.6)	21.48±6.46	04(02.5)	17.21±2.21	04(02.5)	21.88±1.54	01(0.6)	36.94
18.5-24.0	57(35.6)	19.22±7.22	37(23.1)	15.28±3.67	16(10.0)	23.64±2.80	04(2.5)	38.04±2.16
24.1-29.9	72(45.0)	18.68±5.99	44(27.5)	14.93±2.91	25(15.6)	23.21±2.42	03(1.9)	35.88±3.58
≥30	22(13.8)	15.51±6.12	17(10.6)	12.87±3.80	05(03.1)	24.47±2.94	--	--
P value		0.067^{ns}		0.041^{s}		0.472^{ns}		0.629^{ns}
Skin complexion								
Black	35(21.9)	18.43±8.97	23(14.4)	13.47±4.23	07(04.4)	21.79±1.70	05(3.1)	36.55±3.24
Brown	59(36.9)	18.63±5.02	36(22.5)	15.36±2.47	22(13.8)	23.15±1.89	01(0.6)	36.94
Fair	66(41.2)	18.65±6.59	43(26.9)	15.06±3.55	21(13.1)	24.12±3.12	02(1.3)	37.09±2.61
P value		0.987^{ns}		0.099^{ns}		0.094^{ns}		0.729^{ns}

We divide female participants according to use of veil as dressing and 44(27.5%) participants wear veil (Borkha). We took daily approximate sun exposure history as it has a tremendous role in vitamin D metabolism. Fifty-four (33.7%) participants had daily approximate 0-½ hour history of sun exposure, 45(28.1%) had daily approximate ½-1 hour history of sun exposure, 31(19.4%) had 1-2 hours daily approximate sun exposure, and rest 30(18.8%) had daily approximate >2 hours history of sun exposure (Table V).

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Table V: Vitamin D level of study participants according to daily sun exposure (n=160) and outdoor dressing of female (n=91)

Characteristics		25(OH) D Level						
			Deficient		Insufficient		Sufficient	
Sun exposure		Total	≤20ngm/dl		>20ngm/dl-30ng/dl		>30ngm/dl	
0- ½ hour	54(33.7)	14.18±4.36	49(30.6)	13.43±3.83	05(03.1)	21.56±0.48	00(0.0)	--
½-1 hour	45(28.1)	18.42±3.71	28(17.5)	16.02±2.28	17(10.6)	22.38±1.51	00(0.0)	--
1-2 hour	31(19.4)	22.47±7.28	13(08.1)	16.12±2.74	15(09.4)	24.76±2.83	03(1.9)	38.63±0.17
>2 hour	30(18.8)	22.79±7.54	12(07.5)	16.18±2.62	13(08.1)	23.75±2.94	05(3.1)	36.17±
P value		<0.001[s]		0.001[s]		0.015[s]		0.219[ns]
Dressing in female (n=91)								
Veil	44(27.5)	15.21±4.40	40(25.0)	14.46±3.71	04(02.5)	22.73±3.93	00(0.0)	--
Non-veil	47(29.4)	20.11±6.50	24(15.0)	15.11±2.60	21(13.1)	24.07±2.98	02(1.3)	38.53±0.03
P value		<0.001[s]		0.460[ns]		0.438[ns]		--

Significant difference in vitamin D_3 level observed in different duration sun exposed groups (p value <0.001). From Games- Howell Post Hoc test it was observed that 0-½ hour and ½-1 hour sun exposed groups had significantly lower vitamin D_3 level than other groups but between 1-2 hours and >2 hours groups had no significant difference in vitamin D_3 level. Women who used to wear veil had significant (p value <0.001) lower vitamin D_3 level (15.21±4.40ngm/dl) than who did not used to wear veil indicates lack of sun exposure (Table V).

Discussion

This small-scale study done in a district hospital of Bangladesh reveals high rate of vitamin D deficiency, particularly in women. Mean level of vitamin D was 18.60±6.59ngm/dl and total 95.0% (152/160) subjects (deficient 63.7% and insufficient 31.3%) were suffered from deficient or insufficient level of vitamin D. These findings are more of less similar to one recent study of Bangladesh and other study of India, Saudi Arab, Singapore and UK[10,11,12,13,14]. During exposure to sunlight 7-dehydrocholesterol in the skin absorbs UVB radiation and is converted to provitamin D_3which in turn isomerizes into vitamin D_3[15]. We had divided the duration of sun exposure time in four categories but we noticed that only 1-2 hours sun exposure was sufficient for normal vitamin D level which is significant in our study (p=0.001). In comparison to the occupational groups, the farmers used to work mostly in the sunny environment. There was nobody in the farmer group belonged to the actual deficit group and their mean 25(OH)D level was 30.49ngm/dl with a SD of ±7.38ngm/dl, statically significant (p=0.001). The mean value of vitamin D level in the farmer's group was really higher than that of other groups. With the same reason for more sun exposure, the rural group in our study had significantly high level of vitamin D_3 than that of urban group which was similar to other studies[16,17]. Majority of our population in Bangladesh are Muslim where most of the woman used to wear their traditional dress named Borkha (veil). The dress is usually made of silky black cloth which constitutes the strong barrier of adequate sun exposure because the exposed part is only their eyes during wearing the Borkha (veil). The woman is engaged in household occupation and used sun block and umbrella when they go out which are another factor for significant sun exposure barrier. In our recent study, the female subgroup who used the traditional dress Borkha (veil) had significantly lower level of vitamin D than the woman was non veiled[16,18]. It is well established that the people with higher BMI will be lower level of vitamin D. The reason behind the lower level of vitamin D among the person with

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higher BMI is still unclear. It is possibly due to accumulation of vitamin D in the adipose tissue and then it's lower bioavailability in serum level[19,20]. A bi-directional genetic study, which limits confounding, has suggested that higher BMI leads to lower vitamin D, with the effects of lower vitamin D on BMI likely to be small[21]. In addition, the obese people like to engage themselves in sedentary lifestyle and indoor activities, furthermore they avoid sun exposure for easy tiredness and sweaty. The recent study didn't observe significant difference in vitamin D level among the different BMI groups but the obese deficient subjects had significant lower level of vitamin D. In a recent Chinese study there was no association between vitamin D deficiency and BMI[22]. Vitamin D level might be varied according to different age and sex groups. The cause of age-specific sex differences in vitamin D is still unclear, but the reason might be the higher body fat, avoidance of sun exposure, using sunblock cream and covering cloths during going outside[23]. Generally, the older people are more vulnerable to lower vitamin D level due to lower endogenous production of vitamin D and its metabolism in liver and kidney and lack of adequate sun exposure[24,25]. But we didn't find any significant difference in vitamin D level among the different age group as in other study[26]. Usually the people who are highly educated having greater knowledge to practice the healthy behaviors. Though there is observed a good health consciousness among the educated people but there is a negative relationof vitamin D deficiency and education[27]. Possibly the reason behind that, there is more education with more engaged in indoor activities. Another factor may be that the people aren't well concern about the vitamin D deficiency and its effects on healthin our region. Likewise, we observed no significant difference in Vitamin D level among different educational levels in our study. The serum vitamin D level directly depends on the production of vitamin D in the skin under the influence of UVB radiation with wavelengths of 290-315nm. Sun exposure stimulates the production of vitamin D from 7-dehydrocholesterol in the skin. Bangladeshis have mostly type V skin (as defined on the Fitzpatrick scale) which protects against the sun but also reduces absorption of ultraviolet B sunlight that is needed to produce vitamin D by the skin. On the contrary, fair-skinned individuals synthesize more vitamin D when exposed to the same radiation regime[28]. So, the darker skin requires more sunlight to synthesize a given amount of vitamin D. We noticed in our study a more duration (1-2 hours/day) of sun exposure required for optimum level of vitamin D. Unfortunately, we didn't find any significant variation in vitamin D deficiency among the different skin complexion groups which supported other studies[29,30].

Conclusion

With the high rate of vitamin D deficiency in our small-scale study conducting in a district of Bangladesh demands a nationwide survey for vitamin D status.We wish that the media, public health professionals and national health policy maker should pay immediate attention on this "silent epidemics" of hypovitaminosis D and make measurements for fortification of certain commonly used food items with vitamin D.

Limitations

Study conducted in a limited area with a limited number of cases. Randomization in sampling was not done. Dietary habit was not considered. Associated comorbidity was not evaluated extensively.

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